

Data Exploration of ExcessReport

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Project Statement

Explore ExcessReport.txt file and search for interesting patterns.

Data Analysis

1. Load and transform data

```
library(ggplot2)
library(knitr)
f <- read.delim('ExcessReport.txt', sep='\t')
f <- data.frame(f)
f$GrReg7.12less <- as.numeric(f$GrReg7.12less)
f$Hist.6.Mo <- as.numeric(f$Excess.6.Mo.)
f$Hist.6M.CST <- as.numeric(f$Hist.6M.CST)
f$Hist.12.Mo <- as.numeric(f$Hist.12.Mo)
f$Hist.12M.CS <- as.numeric(f$Hist.12M.CS)
names(f)
```

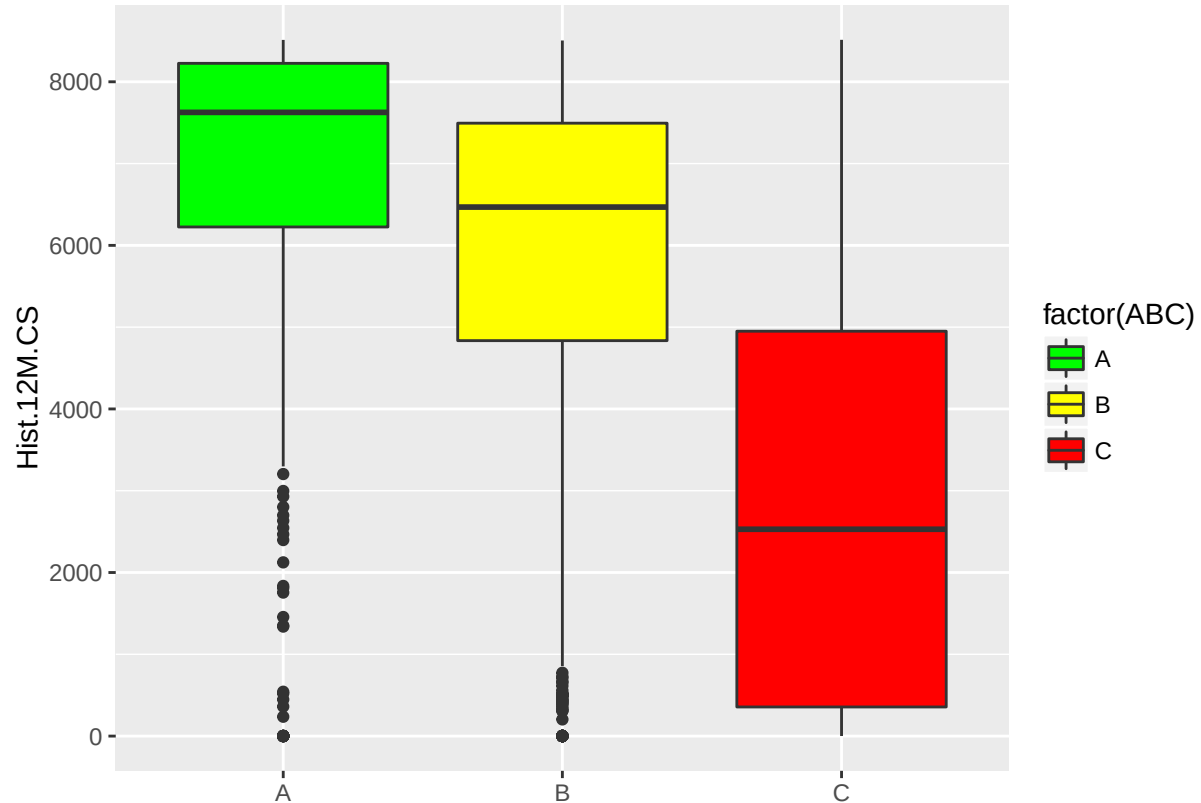
```
## [1] "Item.Num"          "Item.Description"  "Material.Status"
## [4] "OH.Qty"            "OH.Cost"          "Landed.Cost"
## [7] "Excess.6.Mo."      "Excess.6M.C"      "Gr.Reg.6Mo"
## [10] "Gr.Reg.6M.C"       "MOQ"              "Gr.Reg.7Mo"
## [13] "Gr.Reg.7M.C"       "GrReg7.12less"    "Hist.6.Mo"
## [16] "Hist.6M.CST"       "Hist.12.Mo"       "Hist.12M.CS"
## [19] "Merc.Type"         "Product.Hierarchy" "Brand"
## [22] "Category"          "Sub.Category"      "ABC"
## [25] "Ticketed.For"      "MSRP"              "MAP"
## [28] "SAP"               "New.MSRP"          "MSSP"
## [31] "Tckt.Prc"          "Plant"             "Inner.Pack"
## [34] "Master.Pack"       "Catlg.Prc"         "Profit.Ctr"
## [37] "MRP.Controller"    "Sfty.Ind."         "Sfty.wkdays"
## [40] "UPC.."             "Excess.9.Mo."      "Excess.9M.C"
## [43] "Gr.Reg.9Mo"        "Gr.Reg.9M.C"       "Material.Text1"
## [46] "Material.Text2"    "Material.Text3"    "Material.Text4"
## [49] "Material.Text5"    "Material.Text6"    "Material.Text7"
## [52] "XDst.Status"       "Mate"              "X"
```

```
dim(f)
```

```
## [1] 10688    54
```

2. Boxplot of Hist.12M.CS Over ABC

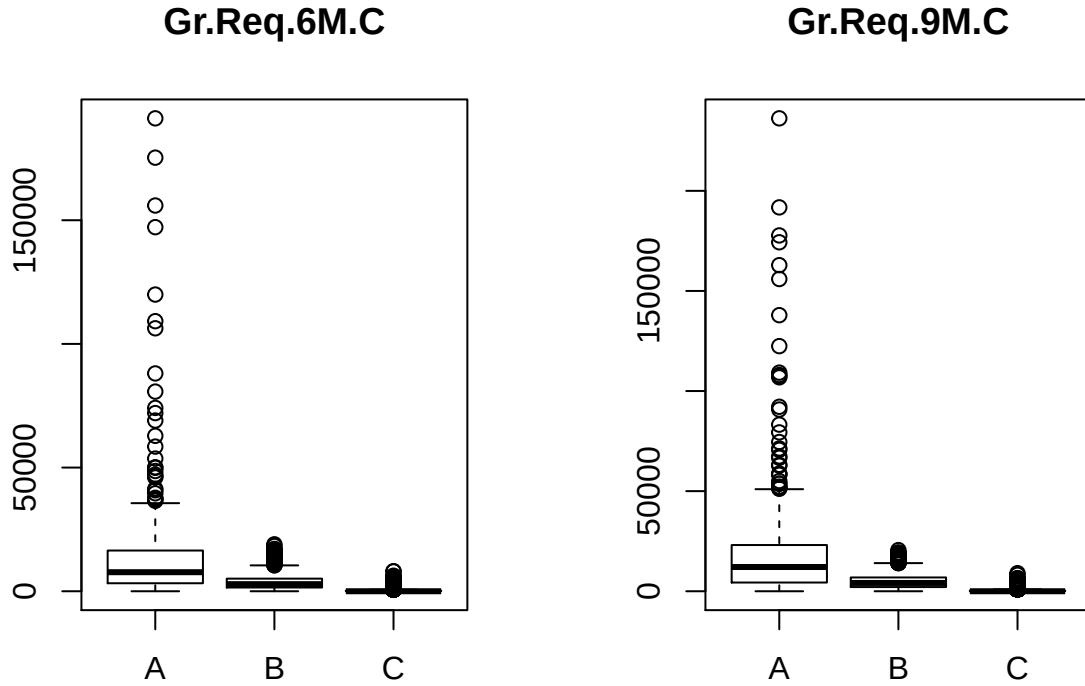
```
ggplot(data=f, aes(x=factor(ABC), y=Hist.12M.CS, fill=factor(ABC))) +
  geom_boxplot() +
  labs(y='Hist.12M.CS', x='') +
  scale_fill_manual(values=c("green", "yellow", "red"))
```



From this graph, we can see on average, Group A and B have higher number of Hist.12M.CS than group C.

3. Boxplot of Gr.Req.6M.C and Gr.Req.9M.C Over ABC

```
par(mfrow=c(1,2))
boxplot(f$Gr.Req.6M.C ~ f$ABC, main='Gr.Req.6M.C')
boxplot(f$Gr.Req.9M.C ~ f$ABC, main='Gr.Req.9M.C')
```



From these two graphs, we can see group C performs poorly in term of Gr.Req.6M.C/9M.C.

4. Check number of items in Group A, B and C

```
f_C <- f[f$ABC=='C',]
f_B <- f[f$ABC=='B',]
f_A <- f[f$ABC=='A',]
Group <- c('A', 'B', 'C')
t_abc <- length(f_A$ABC)+length(f_B$ABC)+length(f_C$ABC)
Number <- c(length(f_A$ABC),length(f_B$ABC),length(f_C$ABC))
Percentage <- round(c(length(f_A$ABC)/t_abc*100,length(f_B$ABC)/t_abc*100,length(f_C$ABC)/t_abc*100))
f_abc <- data.frame(Group=Group, Number=Number,Percentage=Percentage)
kable(f_abc)
```

Group	Number	Percentage
A	524	5
B	1181	11
C	8983	84

From the table, we can see there are 84% of items fell in group C.

5. Check Gr.Req.9M.C numbers in Group C

```
summary(f_C$Gr.Req.9M.C)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0      0.0      0.0   365.0   374.5  9020.0
```

```
length(f_C[f_C$Gr.Req.9M.C==0,]$Item.Num)
```

```
## [1] 4648
```

We can see that 50% of items' Gr.Req.9M.C in Group C fall between 0 and 365, and total 4648 items have Gr.Req.9M.C = 0.

6. Create new column called 'Level_Sale' based on Gr.Req.9M.C

```
f$Level_Sale <- 'low'
f[f$Gr.Req.9M.C>0 & f$Gr.Req.9M.C<300,]$Level_Sale <- 'middle'
f[f$Gr.Req.9M.C>=300,]$Level_Sale <- 'high'
f$Level_Sale <- as.factor(f$Level_Sale)
kable(summary(f$Level_Sale))
```

high	4156
low	4653
middle	1879

7. Build a model to predict items' Level_Sale based on history data

```
# a. Data transformation
f$Hist.6M.Level <- 'high'
f$Hist.12M.Level <- 'high'
f[f$Hist.6M.CS >= 0 & f$Hist.6M.CS < 1500,]$Hist.6M.Level <- 'low'
f[f$Hist.6M.CS >= 1500 & f$Hist.6M.CS < 5000,]$Hist.6M.Level <- 'middle'
f[f$Hist.6M.CS >= 5000,]$Hist.6M.Level <- 'high'
f[f$Hist.12M.CS >= 0 & f$Hist.12M.CS < 1500,]$Hist.12M.Level <- 'low'
f[f$Hist.12M.CS >= 1500 & f$Hist.12M.CS < 5000,]$Hist.12M.Level <- 'middle'
f[f$Hist.12M.CS >= 5000,]$Hist.12M.Level <- 'high'
f_ml <- f[,c(19,32,36,45,53,55,56,57)]
f_ml$Hist.12M.Level <- as.factor(f_ml$Hist.12M.Level)
f_ml$Hist.6M.Level <- as.factor(f_ml$Hist.6M.Level)
str(f_ml)
```

```
## 'data.frame': 10688 obs. of 8 variables:
## $ Merc.Type : Factor w/ 25 levels "", "AFR", "BIN", ...: 19 5 22 13 5 19 20 19 5 5 ...
## $ Plant : Factor w/ 11 levels "1000", "1101", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Profit.Ctr : Factor w/ 18 levels "0000025100", "0000040005", ...: 2 6 10 2 6 3 2 3 8 6 ...
## $ Material.Text1: Factor w/ 4 levels "", "DARK BROWN FINISH FLATWARE CHEST", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Mate : Factor w/ 3 levels "", "ZACC", "ZITM": 3 3 3 3 3 3 3 3 3 3 ...
## $ Level_Sale : Factor w/ 3 levels "high", "low", "middle": 1 2 2 1 2 1 1 1 1 2 ...
## $ Hist.6M.Level : Factor w/ 3 levels "high", "low", "middle": 1 2 1 3 2 1 1 3 1 2 ...
## $ Hist.12M.Level: Factor w/ 3 levels "high", "low", "middle": 1 3 1 3 2 1 1 1 1 2 ...
```

```
summary(f_ml)
```

```
##      Merc.Type      Plant      Profit.Ctr
## OSS      :3225    1000    :6199    0000040005:2964
## OSH      :2655    1500    :2009    0000040025:1375
## BXD      :2253    SYML    :1311    0000025100:1278
##          : 804    1200    : 572    0000040050: 804
## CPT      : 561    1600    : 309    0000040010: 766
## PKG      : 521    1801    : 82     0000040045: 683
## (Other): 669    (Other): 206    (Other)   :2818
##
##              Material.Text1    Mate      Level_Sale
##              :10685           : 1      high :4156
## DARK BROWN FINISH FLATWARE CHEST: 1    ZACC:1094    low  :4653
## JEWELRY BOX W/ DRAWER AND WINDOW: 1    ZITM:9593    middle:1879
## POTTERY BARN MAXINE BRIDAL CAKE : 1
##
##
##
## Hist.6M.Level Hist.12M.Level
## high :2752    high :3527
## low  :4285    low  :3558
## middle:3651    middle:3603
##
##
##
##
```

```
# b. Split data into two sets: training and testing
```

```
library(lattice)
library(ggplot2)
library(caret)
inTrain <- createDataPartition(y=f_ml$Level_Sale, p=0.7,list=FALSE)
training <- f_ml[inTrain,]
testing <- f_ml[-inTrain,]
dim(training)
```

```
## [1] 7484      8
```

```
dim(testing)
```

```
## [1] 3204      8
```

```
table(training$Level_Sale); table(testing$Level_Sale)
```

```
##
##   high    low middle
##   2910    3258   1316

##
##   high    low middle
##   1246    1395    563
```

```
# c. Build a model
library(randomForest)
```

```
## randomForest 4.6-12
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
```

```
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:ggplot2':
```

```
##
```

```
##      margin
```

```
model <- randomForest(Level_Sale ~., data=training)
model
```

```
##
```

```
## Call:
```

```
## randomForest(formula = Level_Sale ~ ., data = training)
```

```
##              Type of random forest: classification
```

```
##              Number of trees: 500
```

```
## No. of variables tried at each split: 2
```

```
##
```

```
##              OOB estimate of  error rate: 32.87%
```

```
## Confusion matrix:
```

```
##              high  low middle class.error
```

```
## high      2494  401      15  0.1429553
```

```
## low       806 2393      59  0.2655003
```

```
## middle    758  421     137  0.8958967
```

```
importance(model)
```

```
##              MeanDecreaseGini
```

```
## Merc.Type      162.5260601
```

```
## Plant          101.0553549
```

```
## Profit.Ctr     202.3867602
```

```
## Material.Text1    0.0682916
```

```
## Mate           44.0757663
```

```
## Hist.6M.Level   560.0981804
```

```
## Hist.12M.Level  366.3360842
```

```
# d. Test the model
```

```
predicted <- predict(model, testing)
```

```
table(predicted)
```

```
## predicted
```

```
##   high   low middle
```

```
##   1743   1377    84
```

```
library(e1071)
confusionMatrix(predicted, testing$Level_Sale)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction high  low middle
##      high   1072  327   344
##      low    166 1044   167
##      middle    8   24    52
##
## Overall Statistics
##
##           Accuracy : 0.6767
##           95% CI : (0.6601, 0.6928)
##      No Information Rate : 0.4354
##      P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.4581
##  McNemar's Test P-Value : < 2.2e-16
##
## Statistics by Class:
##
##           Class: high Class: low Class: middle
## Sensitivity           0.8604      0.7484      0.09236
## Specificity           0.6573      0.8159      0.98788
## Pos Pred Value        0.6150      0.7582      0.61905
## Neg Pred Value        0.8809      0.8079      0.83622
## Prevalence            0.3889      0.4354      0.17572
## Detection Rate        0.3346      0.3258      0.01623
## Detection Prevalence  0.5440      0.4298      0.02622
## Balanced Accuracy      0.7588      0.7822      0.54012
```

The model gave around 67% Accuracy of prediction, which is not bad at all.

Discussion

- *Gr.Req.9M.C* is a good attribute to check the performance of product. And company should avoid the situation like, $Gr.Req.9M.C = 0$. So, it is useful to have a model to predict whether certain product could fall into this kind of situation based on history data.
- Here I used random forest algorithm to build the model, which has 67% accuracy. Based on test result, this model is a good model. Moreover, there is a room to improve the model, such as bringing in more attributes, trying other algorithms, etc.
- Besides *Gr.Req.9M.C*, finding out how much money company earns from each product will be another good way to check the performance. So, next step I will explore the data at this direction.